

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)			burns and produces heat / acts as a fuel (1) forms carbon monoxide (1) carbon monoxide reduces the iron oxide / is a reducing agent (1) award last marking point for coke as a reducing agent if no reference to carbon monoxide	3			3		
	(b)	(i)		100g of CaCO_3 can produce 56g of CaO (1) 8g CaCO_3 can produce $\frac{56}{12.5}$ g of CaO or 4.48g of CaO (1) award (1) for any of following 4.48 to one decimal place is 4.5 therefore Charlie is correct 4.48 to one decimal place is 4.5 4.48 therefore Charlie is correct do not accept 4.5 and Charlie with no working alternative methods 0.08 mol CaCO_3 (1) $0.08 \times 56 = 4.48\text{g CaO}$ (1) award (1) for any of following 4.48 to one decimal place is 4.5 therefore Charlie is correct 4.48 to one decimal place is 4.5 4.48 therefore Charlie is correct			3	3	3	3